Microbe Foundation — Research Status and Tier-1 Submission Roadmap

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Executive summary

This project asks **when protein-set pooling matters for bacterial trait prediction**, validates the mechanism on pathogenicity, and diagnoses **why performance collapses on novel taxonomic families**. The core scientific arc is largely complete in code and manuscript form. What remains for **ICML / NeurIPS / AAAI main track** is to demonstrate that cross-clade improvements hold on the **trained full model** (not only linear probes), add **one strong novelty pillar** (Set Transformer or family-balanced training), and close **reviewer-facing gaps** (pathogenicity confounds, measured localization, figures, more seeds).

Repository: microbe-foundation (main branch, through commit f62607b).

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1. Problem and hypothesis

Setting

- **Data:** 19,592 bacterial genomes, 21 BacDive trait labels, family/genus/species holdout splits.
- **Encoder:** Frozen ESM-2 (esm2_t30_150M, 640-d per protein); ~82M protein embeddings.
- **Variable:** Pooling operator only (mean vs attention vs others in model.py).

Predictability gradient hypothesis

> Attention-pooling helps **machinery** traits (localized gene signal) more than **compositional** traits (diffuse genome-wide signal).

Binding constraint (from paper)

> Cross-clade (family-level) generalization dominates; pooling architecture advantage **vanishes** under family holdout.

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2. Methods (what was built)

2.1 Core model pipeline

- **model.py** — Multi-task MLP on pooled genome vectors; per-protein path with attention/mean/max/top-k/gated pooling.
- **compute_esm2_*.py, embed_from_cache.py** — GPU embedding pipeline (multi-GPU, resumable).

- **Splits** — data/splits.parquet; three-way family split: train / val / test (disjoint families).

2.2 Analysis and monitoring scripts (shipped)

- **microbe_model/monitoring** — Euclidean OOD score vs reference manifold.
- **ood_error_analysis.py** — Table 14 (OOD vs per-genome error).
- **cross_clade_diagnostic.py** — Table 15 + diversity-curve figure.
- **retrieval_head.py** — Table 16 (global probe + k-NN blend).
- **adaptive_retrieval.py** — Table 17 (novelty-ramped blend; negative result).

Leakage control: Probe, k-NN reference, and novelty manifold fit on family-train; blend weights tuned on family-val; family-test scored once.

2.3 Manuscript

Four variants updated with cross-clade section: main, academic, ip_safe, publishable (PDFs via pandoc + tectonic).

2.4 Tests

79 passed on analysis/monitoring suite. Torch tests blocked by local anaconda numpy hang (environment issue, not a code regression).

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3. Results so far

3.1 Main paper — predictability gradient

Split	Compositional ΔF1	Machinery ΔF1	Gap
Species	+0.021 ± 0.002	+0.083 ± 0.012	+0.062
Genus	+0.016 ± 0.004	+0.067 ± 0.010	+0.052
Family	+0.009 ± 0.002	+0.010 ± 0.003	+0.001

Finding: Attention helps machinery traits ~4x more within distribution; advantage collapses at family level.

3.2 Mechanism — pathogenicity (VFDB)

- Top-5 proteins carry ~81% of attention mass.
- ~5x VFDB enrichment vs random proteins in same genome (Wilcoxon $p = 2.5 \times 10^{-7}$).
- Masking top-5 flips ~23% of pathogenic predictions.
- **Caveat:** Taxonomy-majority baseline is strong; matched-clade controls needed for tier-1.

3.3 OOD monitor (Table 14)

- Euclidean k-NN distance: AUROC 0.76 for novel-family detection.
- Diffusion-map variant does not beat Euclidean.
- OOD score is a distribution-shift detector, not a per-genome error oracle.

3.4 Cross-clade diagnostic (Table 15)

Question: Why does family-level transfer collapse?

- **k-NN label transfer:** Matches trained linear probe; AUROC 0.69–0.92; well above chance. Conclusion: **not a representation wall**.
- **Size-controlled diversity curve:** Transfer rises with more training families at fixed genome count. Conclusion: **clade-coverage-limited**.

Traits tested: sporulation, motility, catalase.

3.5 Retrieval-augmented head (Table 16)

Method: $p = \alpha \, p_{\text{probe}} + (1 - \alpha) \, p_{\text{knn}}$, α tuned on family-val.

Trait	α^*	Probe F1	Blend F1	ΔF1	Probe AUROC	Blend AUROC
Sporulation	0.4	0.827	0.844	+0.018	0.925	0.935
Motility	0.4	0.636	0.656	+0.020	0.696	0.721
Catalase	0.4	0.686	0.726	+0.039	0.793	0.797

Finding: Global blend improves cross-clade transfer without encoder fine-tuning.

Scope: Linear probe on mean-pooled 640-d features — **not yet the full model.py system**.

3.6 Novelty-adaptive blend (Table 17) — negative

Trait	$\alpha_{\text{lo}} \to \alpha_{\text{hi}}$	Δ vs global
Sporulation	$0.4 \to 0.4$	+0.000
Motility	$0.4 \to 0.6$	+0.004
Catalase	$0.0 \to 0.4$	-0.016

Finding: Novelty-conditioning does not beat constant $\alpha = 0.4$ (mean ΔF1 vs global = -0.004).

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4. One-paragraph story (current)

Pooling should be chosen by trait localization: attention helps machinery traits in-distribution and is mechanistically grounded for pathogenicity (VFDB). The pooling advantage disappears at family holdout, where the bottleneck is training-clade coverage, not a frozen-embedding wall. Cross-clade signal exists in embedding geometry; a probe + k-NN blend recovers a modest but consistent slice of family-level performance without fine-tuning. Per-genome novelty weighting does not help.

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5. What tier-1 reviewers will ask

Question	Status today
What is new beyond comparing known poolers?	Partial — Set Transformer has no results yet
Does the fix work on the real model?	No — Tables 16–17 use linear probes
Is pathogenicity real or taxonomy?	Incomplete — matched controls uncommitted
Is compositional/machinery split measured?	Weak — eggNOG audit $p = 0.083$
Enough seeds, traits, figures?	Thin — 3 traits; no cross-clade figures in manuscript

Question	Status today
Single encoder scale?	Yes — only 150M ESM-2

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6. What needs to be done (tier-1 main track)

P0 — Submission blockers

6.1 New method or training recipe (at least one; ideally both)

- **Set Transformer pooler (paper section 6):** Implement ISAB + PMA; train 3–5 seeds; all splits including family.
- **Family-balanced training:** Cap genomes per family on family-train; evaluate on family-test.

6.2 Cross-clade on the real system

- Retrieval blend on model.py checkpoint outputs (not LogisticRegression).
- All binary traits; 5+ seeds; compare model vs k-NN vs blend.

6.3 Pathogenicity confound controls

- Matched within-family pathogenic / non-pathogenic controls.
- VFDB enrichment and ablation under controls.

6.4 Measured localization

- Sparse gene-family predictors; correlation with attention gain ($p < 0.05$).

6.5 Figures and statistics

- Cross-clade figure panel; Set Transformer comparison; 5+ seeds with CIs.

P1 — Rebuttal support

- ESM-2 650M re-embed + re-run diagnostics.
- Reproducibility package; fix environment; freeze one manuscript.

P2 — Defer

- Adaptive blend variants (done).
- LoRA fine-tuning before P0.
- Product deploy (not tier-1 main lever).

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7. Go / no-go criteria

Submit ICML / NeurIPS / AAAI main when most are true:

- Set Transformer or family-balanced training shows family-level lift on real model (3+ traits, seeded).
- Retrieval blend beats model-alone on family-test (5+ traits, error bars).
- Pathogenicity survives matched-clade controls.
- Localization measured, not hand-labeled only.
- Cross-clade and method figures included.
- 650M encoder ablation done or bounded.

Fallback if P0 fails: NeurIPS Datasets and Benchmarks, ML4Science workshop, or ISMB / RECOMB.

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8. Suggested execution order

1. Set Transformer implement + train + eval (weeks 1–2)
2. Family-balanced training + retrieval on real model (weeks 2–3)
3. Pathogenicity controls + localization upgrade (week 3)
4. Figures, seeds, 650M, manuscript freeze (week 4)

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9. Venue map

Venue	Fit now	After P0
NeurIPS / ICML main	Weak–borderline	Competitive if method or training lands
AAAI main	Borderline	Competitive with confound controls
NeurIPS D&B	Moderate	Strong as benchmark paper
ISMB / RECOMB	Strong	Ready with polish

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10. Artifacts index

- Main manuscript: paper/predictability_gradient.md / .pdf
- Table 14: paper/tables/14_ood_error_gradient
- Table 15: paper/tables/15_cross_clade_diagnostic
- Table 16: paper/tables/16_retrieval_head
- Table 17: paper/tables/17_adaptive_retrieval
- Figure: paper/figures/cross_clade_diversity_curve.png
- Code: cross_clade_diagnostic.py, retrieval_head.py, adaptive_retrieval.py

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Generated 2026-06-12. Summarizes work through f62607b.